04/16/2026: Coding Workshop: Attention

Reference: *Dive Into Deep Learning*, Chapter 10.

CSCI 546: Diffusion Models

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Course Logistics

Feedback on Reading Survey #8

The original question was: What is attention, and how does it work?

In general, people provided great answers to the first question, but many people struggled (or didn't answer) the second question.

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The easiest way to answer the second question is to reference the notion of keys, queries, and values.

Poll. Would anybody like to give a verbal answer to the second question?

Coding Workshop

Mini-Lecture

Attention as Nadaraya-Watson Kernel Regression

Nonparametric Regression

Suppose we are given n pairs of observations $(x_1, y_1), \ldots, (x_n, y_n)$.

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In non-parametric regression, we want to model the **response variable** Y as a function of **covariates** X making a few assumptions as possible (e.g. without assuming linearity).

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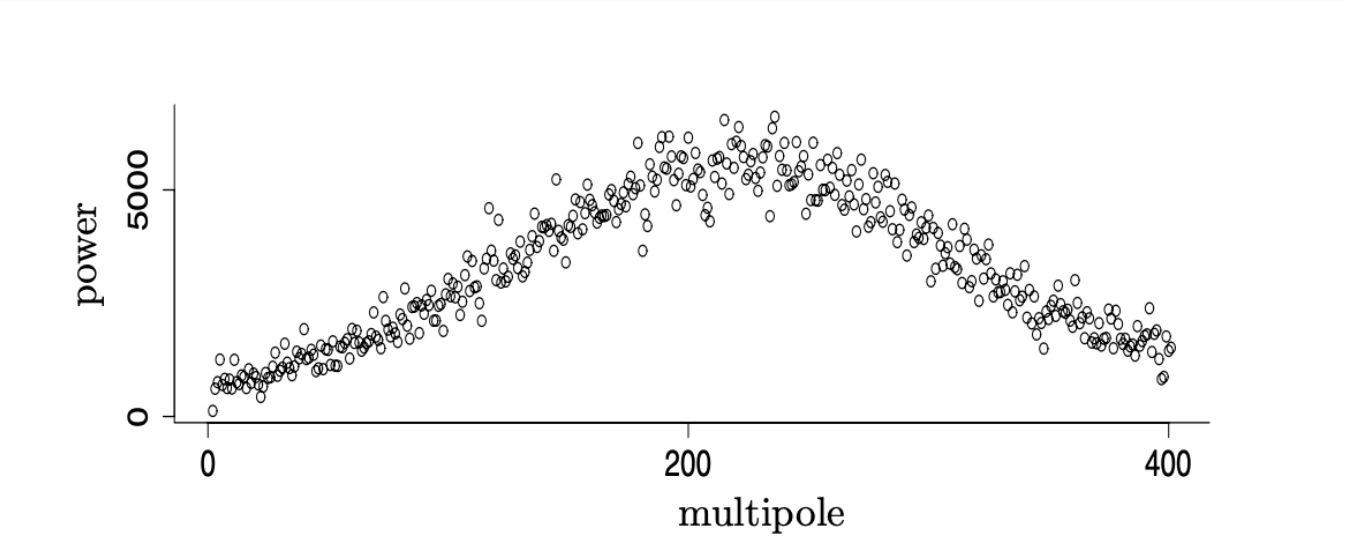
In non-parametric regression, we want to model the **response variable** Y as a function of **covariates** X making a few assumptions as possible (e.g. without assuming linearity).

This allows us to perform regressions for datasets such as the Cosmic Microwave Background (CMB) radiation:



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Nonparametric Regression

We assume the **response variable** Y is related to known **covariates** $X = x$ by the equation

$$Y_i = r(x_i) + \epsilon_i, \qquad \mathbb{E}[\epsilon_i] = 0, \quad i = 1, \ldots, n.$$

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$$Y_i = r(x_i) + \epsilon_i, \qquad \mathbb{E}[\epsilon_i] = 0, \quad i = 1, \ldots, n.$$

We may interpret the "trend" $r(x)$ as the mean of Y conditional on $X = x$, that is

$$r(x) = \mathbb{E}[Y \mid X = x].$$

Smoothing Kernel

Smoothing kernels are for taking local averages.

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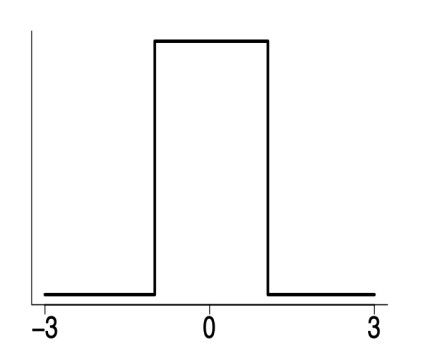
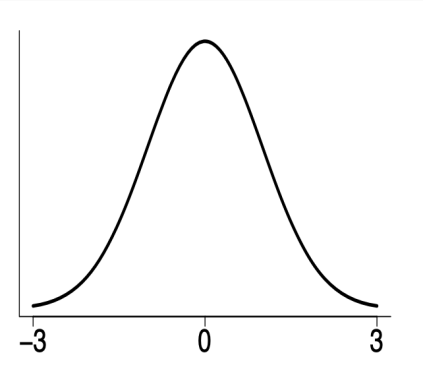
Smoothing Kernel

Smoothing kernels are for taking local averages. Examples:

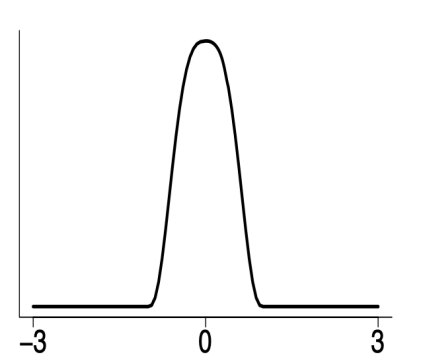


Boxcar



Gaussian

Epanechnikov



Tricube

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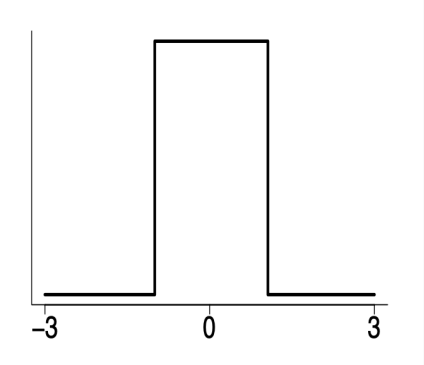
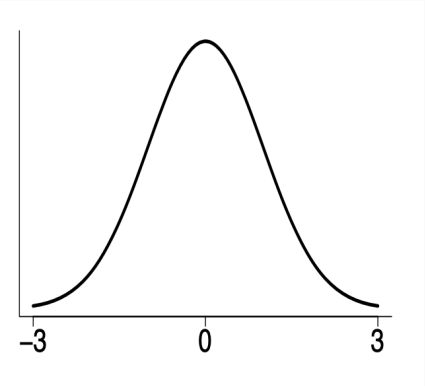
Smoothing Kernel



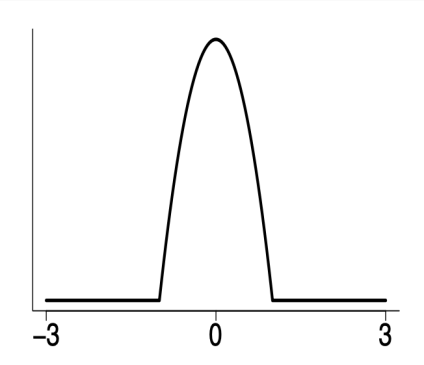
Boxcar

$K(x) = \frac{1}{2}\, I(x)$



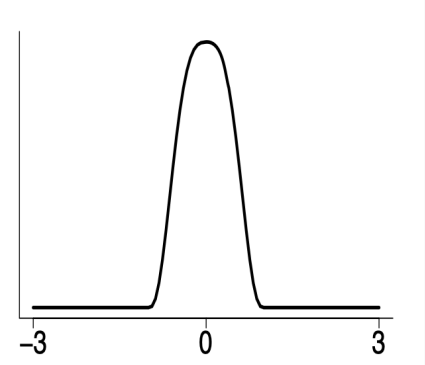
Gaussian

$K(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$



Epanechnikov

$K(x) = \frac{3}{4}(1 - x^2)\, I(x)$



Tricube

$K(x) = \frac{70}{81}(1 - |x|^3)^3\, I(x)$

$$I(x) = \begin{cases} 1, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$

Smoothing Kernel

Definition

A **smoothing kernel** is any smooth function K such that $K(x) > 0$ and

$$\int K(x)\mathrm{d}x = 1, \qquad \int x\,K(x)\mathrm{d}x = 0, \quad \text{and} \quad \sigma_K^2 := \int x^2 K(x)\mathrm{d}x > 0$$

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Poll. What's an easy way to construct a smoothing kernel?

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Poll. What's an easy way to construct a smoothing kernel?

Solution. Just take the p.d.f. of a (non-degenerate) random variable with mean 0.

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Estimating the trend

Definition

The **Nadaraya-Watson kernel estimator** is defined by

$$\widehat{r}(x) = \sum_{i=1}^{n} w_i(x) y_i$$

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where the weights $w_i(x)$ are given by

$$w_i(x) = \frac{K(\frac{x-x_i}{h})}{\sum_{j=1}^{n} K(\frac{x-x_j}{h})} \tag{1}$$

for a smoothing kernel K with **bandwidth** $h > 0$.

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Poll. Who can explain what's going on here?

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Remark. Basically, we estimate the trend $r(x)$ by taking a weighted average of the observations (y_i)'s, giving higher weight to the observations that are "close" to x.

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Remark. Basically, we estimate the trend $r(x)$ by taking a weighted average of the observations (y_i)'s, giving higher weight to the observations that are "close" to x. The larger the bandwidth h, the more observations that are considered "close".

Example: CMB Data

Let's use Nadayara-Watson kernel regression to estimate the trend for CMB data.

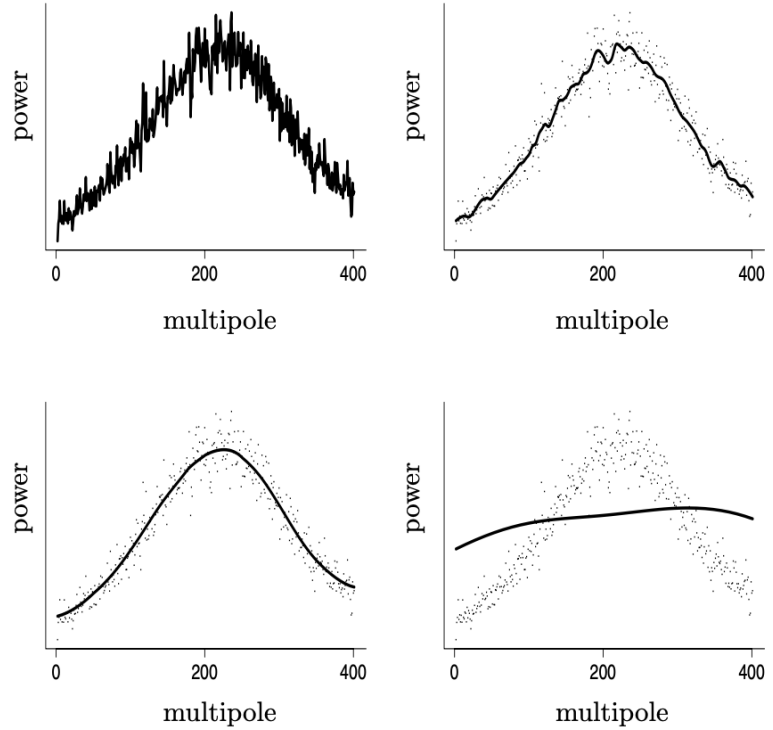

FIGURE 5.2. Four kernel regressions for the CMB data using just the first 400 data points. The bandwidths used were $h = 1$ (top left), $h = 10$ (top right), $h = 50$ (bottom left), $h = 200$ (bottom right). As the bandwidth h increases, the estimated function goes from being too rough to too smooth.

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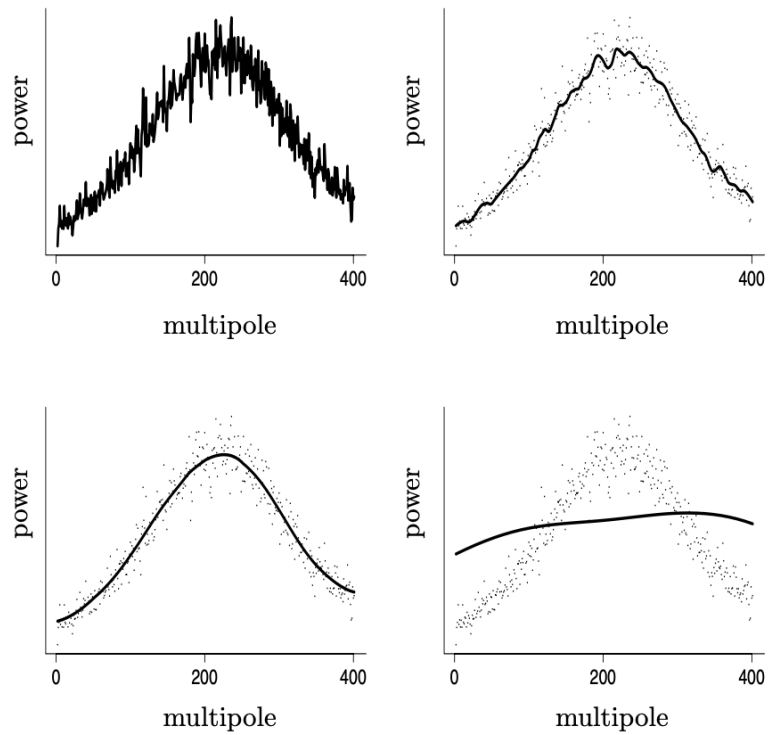

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Remark. The optimal bandwidth can be chosen via cross-validation.

Discussion: How is attention pooling related to Nadayara-Watson kernel regression?

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Attention as Nadaraya-Watson

Observation

Attention can be expressed as

$$\underbrace{\Delta y_j}_{\text{change in embedding}} = \sum_{i=1}^{n} \underbrace{w_j(k_i)}_{\text{weight}} \underbrace{v_i}_{\text{value (for } y_i)}$$

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Poll. How is attention pooling related to Nadayara-Watson kernel regression?

- This kernel is learned rather than fixed in advance.
- Each unit j gets its own kernel.

...